This comprehensive article serves as a deep synthesis of the source documents, detailing the architecture, functionality, integration protocols, and device-specific configurations for the NXGEN Genesis platform and its associated systems, primarily Evalink Talos.

---

# The NXGEN Knowledge Repository: Genesis, Talos, and Unified Security Management

## I. Platform Foundation: Genesis and Talos Overview

### A. NXGEN Genesis: A Cloud-Based USMS Platform

**Genesis is a cloud-based Software as a Service (SaaS) platform** providing video surveillance and Internet of Things (IoT) services. It functions as a **video-centric Unified Security Management Service (USMS)** and a cloud-based video alarm handling platform. The platform enables users to manage and monitor security devices remotely via a web browser or mobile application, thereby **eliminating the need for on-premise hardware and software management**.

**Core Benefits and Value Propositions of Genesis:**

1. **High Performance and Efficiency:** Genesis is designed for enhanced efficiency and a user-friendly experience.
2. **False Alarm Reduction:** The platform integrates with existing infrastructure and leverages **AI analytics to significantly reduce false alarms**, potentially by up to **80%**. It uses advanced algorithms to detect human or vehicle activity, distinguishing real threats from false triggers like wind or animals. Critical alerts are prioritized and may be handled within 60 to 90 seconds.
3. **Robust Cloud Architecture:** Genesis utilizes a secure, scalable, and accessible cloud infrastructure.
4. **Multi-Tenant Support:** The platform supports a **multi-tenant architecture**, serving multiple customers through subdomains under `inexchange.cloud` (e.g., `ccontact.inexchange.cloud`) or `nxgen.cloud`.
5. **Centralized Management and Hierarchy:** Genesis uses a **tenant-based model** for organizing customer data. Devices are managed within a **hierarchical structure**: **Tenant → Customer → Site → Device → Cameras/Sensors**.

## B. Evalink Talos: Alarm Management and Workflows

**Evalink Talos** is a specialized, automated, cloud-based platform for monitoring stations designed to manage security alarms. It enhances the efficiency of security operations through advanced alarm management.

**Key Aspects of Talos:**

- **Architecture:** Talos is highly reliable, built using Amazon Web Services (AWS) servers across America, Europe, and Asia in different availability zones. It is composed of many small, independent **microservices**.
- **Role in Alarm Flow:** Alarms generated by devices often flow to Genesis first, where they are processed with analytics, and the outcome ("real alarm" or "false alarm") is then sent back to Talos as a **follow-up alarm** for enriched and verified event data.
- **Site Synchronization:** When a site is created in Genesis, the site name is **automatically created in Talos**. Site data synchronization has also been implemented.
- **Alarm Definition:** An alarm is defined as a single notification from a security device (e.g., motion detected by a camera) requiring attention.
- **Interface Screens:** Operators interact with Talos via three main screens, typically on multiple monitors: the **Alarm Receiving Screen** (displays incoming alarms, sortable by priority or time), the **Video Screen** (shows live or recorded video clips linked to alarms), and presumably a third dashboard/overview screen.
- **Operator Roles:** Roles are typically segregated: the **Night Shift** focuses on critical alarms (intrusion, fire), deferring technical issues. The **Day Shift** handles technical alarms, maintenance, workflow adjustments, and customer interaction.

# II. Technical Architecture and Infrastructure

## A. Microservices and Proxy Architecture

Genesis employs a **proxy architecture** utilizing specialized microservices to handle device-specific communications. This design ensures that the core Genesis system interacts uniformly with devices through standardized Application Programming Interfaces (APIs).

- **Proxy Function:** There is typically a **proxy per device type** (e.g., Dahua proxy, Hikvision proxy, Hanwha proxy, Axxon proxy). The proxy's functionality is device-specific, even though the input signature and output response schema remain **common** across all device types.
- **Communication Flow (UI/API to Proxy):**
    1. A user (web or mobile) initiates a request (e.g., PTZ control).
    2. The request goes to the **API**, which is the **only hop for communication** for the front end.
    3. The API identifies the request type and the device type (e.g., PTZ for an Hikvision device).

4. The API consults a **dictionary of proxy endpoints** (e.g., `hikproxy.nx.cloud`).
5. The API, acting as a "middleman" or "redirector," makes an **HTTP call** to the respective proxy with a standardized **JSON-based signature**.
6. The proxy handles the actual processing based on the device's specific protocol (SDK, TCP, REST, etc.).
7. The proxy responds back to the API with a common schema, and the API passes it to the user interface.

## B. Device Protocol Flavors

The microservice architecture is necessary because devices utilize multiple protocol flavors:

| Protocol Type | Usage Examples | Notes |
|---|---|---|
| **HTTP/REST/Open API** | Used for basic request/response functions like getting device information, camera lists, or PTZ control (for Hikvision/Hik-Connect Pro). | REST is not suitable for streaming modes. |
| **SDK** | Required for complex functions like live stream, playback, downloading video files, searching video files, and two-way audio (e.g., Hik-Connect Pro/Hikvision use C# based SDK or .NET HcNetSDK). | SDK-based integration often provides a TCP layer for streaming. |
| **TCP/Native TCP** | Used by older/legacy devices or for specific alarm transmission protocols. ADPRO pushes events via TCP to the Genesis ADPRO Receiver. | |
| **Webhooks/HTTP** | Axis Pushes events via HTTP/WebHook to the Genesis proxy. Hanwha configuration involves adding **HTTP requests** (webhooks) to the camera rule to push events. | |
| **Websocket Subscription** | Axxon events are pulled by Genesis via a websocket subscription. | |
| **SIA DC-09** | Used for alarm transmission from devices like Reconeyez, typically routed through Talos/Virtual SIA DC-09 Receiver. | Requires receiver IP address, port, and account ID of each site. |

## C. Infrastructure and Scaling

The NXGEN system runs on a Kubernetes cluster.

- **Development Environment:** The development environment is separate from production, running on a separate VPC with minimal nodes (2 to 3 nodes).
- **Autoscaling (HPA):** The primary scaling mechanism is **Horizontal Pod Autoscaling (HPA)**, which scales horizontally by adding nodes, rather than Vertical Pod Scaling, which risks service interruption during node upgrades/downgrades.
    1. Autoscaling is crucial for **resource-consumable services**.
    2. Examples requiring autoscaling include the **CVIDR (AI engine)** which detects events and processes images, and also streaming services, Adpro proxy, and event services.
    3. The scaling threshold is set at **75% CPU utilization**. This threshold is chosen to ensure a new node can be initialized and start accepting calls (a 2-3 minute process) before the existing pod reaches 100% utilization.
- **Deployment Workflow (Example: API):**
    1. The base image is a **Node.js-based image**.
    2. `package.json` is copied, and `npm install` runs to install modules.
    3. All code is copied from Git to the container.
    4. `npm run start` is executed to run the code.
    5. The container is built (first stage).
    6. The second stage involves pushing the image to the Docker registry, with a shell script handling tagging parameters (version, commit, branch).

## D. Network and Ports

Various network ports are required for Genesis communication:

- **Genesis Cloud Receivers:**
    - **Genesis Adpro Receiver IP Address** (Specific to tenant) on **Port 10000** for event transmission.
    - **Heitel Gateway 1** (Live Video): 3.123.206.197.
    - **Heitel Gateway 2** (Events): 3.124.38.48.
    - **NTP Server Address:** `timel.nxgen.cloud` is used for time synchronization for Hikvision and Dahua devices.
    - **Twilio SIP URL:** `genesisaudio.sip.twilio.com` on Port 443 (HTTPS) for Genesis Audio.
- **Customer Site Requirements (Examples):**
    - **ADPRO:** RTSP Port 554, Control Port 2000, Audio Port 3000.
    - **Dahua:** Control Port 37777 (TCP), RTSP Port 554 (TCP/UDP), HTTP Port 80, HTTPS Port 443. Also uses port ranges 10000-10500 (TCP) and 14890-14990 (TCP) for Websockets (P2P streaming).
    - **Hikvision:** Control Port 8000, HTTP/HTTPS Port 80/443, RTSP Port 554.

- **Milestone:** Management Client Port 443, Streaming Port 7563, Web Client Port 8081, Event Port 22331.
- **Heitel:** Port 3333 is critical for communication via HTConnect.

# III. Centralized Management and Configuration

## A. Standard Device Onboarding Process in Genesis

Integrating a device involves a standardized sequence within the Genesis Configuration App:

1. **Login to Genesis** and navigate to the **Configuration App**.
2. Add **Customer** and **Site** (if not already configured). Site addition automatically syncs the site name with Talos.
3. Navigate to the **Devices tab** at the site level and click **"Add"**.
4. **Select Device Type** (e.g., HanwhaNVR, Hikvision, Milestone, Teltonika-IOT).
5. **Enter Mandatory Details:** This typically includes **Name**, **IP Address/Host**, **Username**, and **Password**. Specific devices require ports (RTSP, Control, HTTP/HTTPS) and unique identifiers (Serial Number, Server Unit ID, or Base URL).
6. **Time Zone Synchronization:** The device's time zone must be correctly configured in Genesis for accurate real-time event forwarding and synchronization.
7. **Click "Discover":** Genesis initiates an automatic discovery process to connect, fetch device-specific information (including **Input/Output (I/O) details**), and automatically add associated **sensors** (cameras).
8. **Click "Save"** to finalize the configuration.

## B. Custom Properties

Custom parameters provide a free-flow configuration enabled on the fly for different features at various hierarchical levels (Tenant/Customer/Site/Device/Cameras). They are added under "Additional Settings" > "Custom property" after selecting "edit" for the relevant level.

**Key Uses of Custom Properties:**

- **Event Clip Recording:** Used for ADPRO, Dahua, Hikvision, and Milestone to enable automatic recording of **pre- and post-alarm video clips** (e.g., -5 to +5 seconds) for specific real alarms, providing rich context for operators.
  - `Parameter name: eventClipRecord` (Value: True).
  - `Parameter name: eventClipRecordAlarmCode` (Value: e.g., `motion.perimeter`).
- **iFT Gateway (ADPRO):** Required for ADPRO device subtype [iFT] Gateway.
- **Milestone Optimization:** Used to set base URLs, WebSocket endpoints, optimize for low bandwidth, and control alarm/event subscriptions.
  - `MilestoneNVR_Device_Custom_isLowBandwidthDevice` (Value: True).

- **Teltonika Event Filtering:** Used to define **Custom Alarm Rules** with thresholds (e.g., voltage high/low limits) to filter frequent data changes and ensure only meaningful events trigger alarms.
- **HoneywellS35:** Requires `HoneywellS35_Device_Custom_baseUrl` at the service provider level.

### C. Health Monitoring, Site Pulse, and Event Polling

**Device Health Monitoring** is a core capability, providing automated alerts for offline cameras and monitoring storage or battery status.

- **Site Pulse / Lifecheck (Recommended):** This feature acts as a "heartbeat" for the system. The ADPRO device (for example) sends a regular "I am alive" heartbeat signal to the configured Genesis IP address at predefined intervals (e.g., 10 minutes, 30 minutes, or 1 hour).
    - **Timeout Alarms:** If Genesis does not receive a site pulse within the configured time, a **timeout alarm** is automatically raised to the operator in Talos, signaling potential issues like power failure or network outage.
    - Site Pulse Configuration is also supported for Axis IP Camera, Genesis Audio, Milestone, and Hanwha.
- **Event Polling:** This configuration should **only be activated if the device cannot send events directly to Genesis**. In this case, Genesis will poll for events at a customizable time interval. This is critical for devices like Teltonika, which relies on event polling for dashboard data.
- **Alarm Supervision (Talos):** This feature can be used for devices without a constant heartbeat or polling connection. If Talos doesn't receive an expected check-in signal (e.g., a "Fridge OK signal" or a QR code scan by a guard) within a set time, it raises a "Timeout Alarm".

## IV. Alarm Management and Workflow Ecosystem

### A. Talos Workflows

Workflows, managed within the Talos Workflow module, are sets of predefined steps or scenarios that guide how a specific type of alarm is handled. They ensure alarms are processed consistently and efficiently.

- **Types:** Workflows can be **manual** (guiding a human operator step-by-step) or fully **automated** (performing actions without human interaction).
- **Levels:** Workflows can be:
    - **Global:** Apply to all sites.
    - **Group:** Apply to specific site groups.
    - **Site-Specific**.

    - **Managed:** Created once and assigned to multiple sites for easier maintenance and scalability. **Managed workflows are generally recommended**.
- **Structure and Conditions:** A workflow consists of incoming conditions, settings, and steps. Incoming conditions dictate which alarm is assigned to the workflow, based on factors like alarm code, zone, partition, alarm value (Alarm or Restore), or a regular expression in the payload.
- **Steps and Actions:** Steps define actions like sending emails/SMS/calls, checking device status, or requiring operator decisions (e.g., "Is the alarm verified? Yes/No").
- **Customization and Optimization:** Workflows should be customized based on customer operating procedures, such as removing audio announcements for sites without loudspeakers. Workflows can automatically disarm a site when it is set to "test mode" in Talos, which conserves processing resources on Genesis.

## B. User Roles and Access Control

User management in Genesis/Talos is controlled by roles and workspaces.

- **Roles:** Roles define permissions based on APP, category, action, and API. Common roles include Admin, Customer Admin, Backend API QA, and Operator.
- **Workspaces:** Workspaces route alarms and restrict which operators can handle certain alarms (e.g., Standard vs. VIP workspace). A user can be restricted to specific work groups, site groups, or site types.
- **Auto-feed:** Alarms can be forced to be automatically assigned to a user (auto-feed) and prevented from being blocked, depending on customer preference.

## C. Reporting and Audit Logs

Genesis and Talos provide robust reporting capabilities for auditing, accountability, and performance analysis.

- **Report Templates:** Default templates include:
    - **Alarm Listing:** Lists all alarms within a period (time, type, location).
    - **Workflow Reports:** Details the execution of automated workflows, including steps, outcomes, and operator comments.
    - **Event Log:** Records all system events, including alarms, arm/disarm actions, and system messages.
    - **Site Documentation:** Provides comprehensive information about each monitored site, including configuration and status.
- **Customization:** Templates can be customized to add/remove fields, change the layout (requiring Angular-based code knowledge for advanced edits), apply filters, and integrate with other data sources.
- **Audit Logs:** The **Audit tab** in Genesis maintains a detailed record of all actions taken within the tenant (who, type, status, date/time), serving as a crucial audit trail for accountability and troubleshooting.

# V. Device Integration Specifics and Functionalities

Genesis supports integration with a wide array of video and IoT systems, including Hikvision, Dahua, NX Witness, Hanwha, Avigilon, Axis, Milestone, Camect, InnoVi, Reconeyez, Teltonika, ADPRO, and Heitel.

## A. ADPRO (XT/iFT Gateway)

- **Integration Method:** Events are pushed via **TCP** to the Genesis ADPRO Receiver.
- **Required Configuration:** Receiver IP address, port (10000), and the account ID of each site are critical information needed for the DC9 integration.
- **Alarm Behavior:** The system's behavior when Armed or Disarmed can be defined, including options like suppressing alarms, primary/backup transmissions, and email notifications.
- **iFT Gateway Subtype:** For this subtype, specific **custom properties** must be added in Genesis under "Additional Settings" -> "Custom property":
    - `Parameter name: eventClipRecord` (Type: String, Value: True).
    - `Parameter name: eventClipRecordAlarmCode` (Type: String, Value: `motion.perimeter`).
- **Known Issues:** The Server Unit ID is used for device identification, but it is an unreliable integer that can be easily duplicated. Alarms from inactive/disarmed devices will still be forwarded. Time-based alarm image pulling is not available.

## B. Hikvision and HikProConnect

- **Integration Protocol:** Uses **Http ISAPI** for Cloud mode and **.NET HcNetSDK** for Local mode. Event integration utilizes an **ISAPI Event API Subscription**, meaning no specific device configuration is needed on the Hikvision side beyond enabling "Notify Surveillance Center".
- **Time Synchronization:** Configure the NTP Client on the NVR to use **`timel.nxgen.cloud`** as the server.
- **HikProConnect (Cloud):** Requires the **Device Base URL** (e.g., `https://ieu.hik-partner.com` for Europe) in the "IP address-host" field, and the **Encryption Key** must be added. Encryption functionality is only supported when **Local Mode is installed**.
- **Supported Features (Cloud Mode):** Discovery, Live Video, Playback, Timeline, Events, PTZ/Presets control, IO, Genesis Audio (SIP), Arm/Disarm capabilities.
- **Optimization/Troubleshooting:**
    - **Avoid basic motion detection** for alarm reception; instead use smart events like line crossing or intrusion detection.
    - Enabling **Smart Codec** may nullify certain network camera functions.
    - A 5-minute live view limit occurs if **UPnP or manual port forwarding is not enabled**.

    - Ensure the **Server Unit ID is unique** to prevent alarm conflicts due to duplication.

## C. Dahua and Dahua DoLynk

- **Prerequisites (DoLynk):** Installer must provide **Serial Number**, **Device Password**, **User Name**, and **Category code**. For sharing with NXGEN, the email `dev@nxgen.info` (NXGEN Technology AG) must be used in the Company Name/email field within the DoLynk Care account.
- **Supported Modes:** Supports both **Cloud Mode** (integration with Dahua DoLynk Cloud) and **Local Mode**.
- **Local Mode:** Local mode operations are supported for advanced functionalities like **P2P streaming, two-way audio, and encrypted streams** on operator workstations. Local Mode requires prerequisites like Microsoft Redistributables, .NET Framework 4.8, and NodeJS (V18 preferred).
- **Supported Features:** Live Stream, Playback, Events/Alarms (with analytics), PTZ Control, I/O Management.
- **Time Synchronization:** Resolution for incorrect time settings is to use `timel.nxgen.cloud` as the NTP server with an interval set appropriately (e.g., 3600 seconds).

## D. Hanwha-Techwin / NX Witness

- **Configuration:** For event reception, webhooks must be configured inside the camera rules on the Hanwha device.
    - The **Hanwha proxy URL** (`https://hanvoproxy.nxgen.cloud/eventIngest`) must be added.
    - The payload must include **Camera ID** and **event name**.
    - Login/Password for basic authentication should be the device ID.
- **Supported Features (Cloud Mode):** Discovery, Live Video, Playback, Timeline, ARM/DISARM, Genesis Audio (SIP).
- **Limitations: PTZ/Presets and I/O functionalities are not supported in Cloud Mode**. **No Local Mode Support**. Event log pulling is generally not available.

## E. Milestone VMS

- **Required Ports:** Management Client Port 443, Streaming Port 7563, Web Client Port 8081, Event Port 22331.
- **Streaming Protocol:** Uses TCP JPEG and Raw formats (RTSP is not supported).
- **Event Integration:** Utilizes the Milestone **MIP Component SDK** for event integration.
- **Supported Features (Cloud Mode):** Discovery, Live Video, Playback, Timeline, Events, ARM/DISARM, Genesis Audio (SIP), PTZ/Presets, IO.
- **Health Support:** Basic and Basic+ Camera Health are supported, but Advanced Camera Health is **not** supported for Milestone.

- **I/O Management:** Genesis automatically reads and displays I/O configuration details once the device is discovered.

## F. Axxon VMS

- **Event Integration:** Events are **pulled by Genesis** via a **websocket subscription**; no manual configuration is required on the Axxon side.
- **Supported Features (Cloud Mode):** Discovery, Live Video, Playback, Timeline, Events, ARM/DISARM (software arm), Genesis Audio (SIP), PTZ/Presets.
- **Limitation: I/O functionalities are NOT supported** for Axxon VMS.
- **Known Issues:** Passwords might not support special characters such as `@` or `#`. The current alarm polling mechanism is noted as **not reliable**.

## G. Camect AI Video Recorders

- **Live Video:** Provides live view via an **HLS URL**.
- **Supported Features (Cloud Mode):** Discovery, Live Video, Events, ARM/DISARM, Genesis Audio (SIP), PTZ/Presets, Event Acknowledgement.
- **Limitations:** Playback and Timeline features are **not supported by the Camect device itself** in Cloud Mode. Playback is also not supported in Local Mode.
- **Configuration Detail:** Requires a unique HLS stream for each camera to be presented to the monitoring station technical team.

## H. Heitel (Legacy Device)

- **Nature:** Considered a **legacy device**, often requiring manual intervention for successful alarm reception.
- **Onboarding:** Requires **raising an individual ticket** (per device) in the NXGEN helpdesk portal to enable alarm reception due to its legacy nature.
- **Prerequisite:** The device must be reachable via **HTConnect**.
- **Alarm Transmission:** Requires whitelisting Heitel Gateway IPs (3.123.206.197 for video and 3.124.38.48 for events) on the customer's firewall, and Port 3333 must be open.

## I. Reconeyez PIR Cam

- **Primary Integration Flow (Current):**
    1. End customer configures alarms and schedules in their Reconeyez portal.
    2. Customer informs Genesis/Reconeyez, which starts pushing images to a **Genesis FTP**.
    3. Alarms (raw events) are sent from Reconeyez to **Talos via SIA DC-09**.
    4. Talos uses a workflow to convert the C-alarm message to a Genesis alarm.
    5. Genesis receives the alarm, pulls its equivalent images from FTP, **applies analytics**, and sends a **follow-up alarm** back to Talos.

- **Future State:** The goal is to evolve the system so that customers can configure the IP address and port number directly in Genesis, eliminating the need for workflow conversion/alarm forwarding from Talos to Genesis.

### J. Teltonika IoT

- **Primary Purpose:** Teltonika devices (IoT routers/sensors) are integrated primarily for **dashboard data and event polling**, **not for streaming video or audio**.
- **Data Points:** Provides insights into metrics like **battery voltage, charger state, and daily yield**.
- **Critical Filtering:** Due to high frequency of data changes (e.g., voltage changes every second), Genesis uses **Custom Alarm Rules** (added as additional properties) to set specific thresholds (e.g., voltage low: 12.7V). Only events violating these rules are processed as alarms.

### K. Genesis Audio (SIP Twilio)

- **Purpose:** A specialized device type developed by NXGEN to facilitate **remote announcements** and enable **two-way audio communication** through integrated IP speakers.
- **Operation:** Engineered to initiate calls even if the **local mode is disabled** on the device, highlighting its cloud-centric model. Audio functionality is enabled via **SIP (Session Initiation Protocol)**.
- **Configuration:** Requires selecting "Genesis Audio" as the device type and configuring the speaker's SIP URL, username, and password. The audio toggle must be verified as enabled for both the Genesis Audio device and its associated sensors.

# VI. Troubleshooting and Support

General troubleshooting involves a systematic approach across layers, typically starting with device logs, followed by network configuration, and escalating to support if necessary.

| Issue Category | Common Causes | Resolution Steps |
|---|---|---|
| **Alarm Transmission Failure** | Device/Sensor Disarmed; Incorrect alarm profile/schedule configuration; Incorrect Hub ID/Server Unit ID. | Check device dashboard logs (video vs. technical alarms); Verify device arm/disarm schedule alignment; Check Talos receiver status (SIA DC-09 is operational); Ensure the unique Server Unit ID is correct. |

| | | |
|---|---|---|
| **Connectivity/Video Failure** | Ports not open on firewall; Incorrect IP address/DNS; Network instability/bandwidth issues. | Verify all required ports are open and forwarded (check for green status in Genesis); Ensure Genesis service IPs are whitelisted on customer firewall; Check network connections and compatibility. |
| **Time Synchronization** | Device time does not match Genesis time; Incorrect NTP server address. | Ensure NTP server is set to `timel.nxgen.cloud`; Use the Genesis "time zone sync" button if available; Ensure time zone configuration in Genesis matches the physical device. |
| **SDK/EXE Issues** | EXE not running or listening on the correct port. | Ensure the SDK executable is running and monitoring the correct port. |
| **Cloud API Issues** | Invalid authentication tokens or API keys. | Verify and refresh authentication tokens/API keys used by the proxy. |
| **Device Discovery Failure** | Incorrect mandatory fields (e.g., Serial Number, Device Base URL, API Token). | Double-check all entered credentials and required identifiers. If persistent, raise a ticket to the NXGEN support team. |

---

## Analogy for System Integration

The Genesis platform and its device ecosystem function like a central global switchboard that manages diverse communication devices. Each device (Hikvision, Dahua, ADPRO) speaks a different "language" (protocol: REST, SDK, TCP). Instead of requiring the central switchboard (Genesis) to learn every language, specialized **Proxy microservices** act as **universal translators**. A request from the user interface goes to the central API (the receptionist), which directs it to the correct translator (the proxy). The translator figures out the specific native language/protocol needed (SDK, port 8000, etc.) to talk to the device, handles the conversation, and then replies back to the central switchboard in a **standardized, common language (JSON/HTTP)**. This system allows the core functions (like analytics and alarming in Talos) to operate uniformly, regardless of whether the source device is a modern IP camera or a legacy NVR.

This comprehensive article serves as a deep synthesis of the provided source documents, compiling detailed technical information, architecture specifications, device-specific configurations, operational workflows, and support protocols for the **NXGEN Genesis** platform, its associated **Evalink Talos** system, and its expansive device ecosystem.

---

# The NXGEN Knowledge Repository: Genesis, Talos, and Unified Security Management (Continued)

## VI. Support, Troubleshooting, and Knowledge Management

Effective operation of the NXGEN Genesis ecosystem relies on clear support hierarchies, defined troubleshooting procedures, and robust documentation practices.

### A. Support Structure and Ticket Handling

Issue resolution within the NXGEN framework is primarily managed through a structured system, beginning with L1 support.

1. **L1 Support Responsibilities (First Check):** The initial step for L1 support is to determine if **all necessary information is available** to begin troubleshooting the particular issue. This required information includes checking all three parts of the ticket: the **title**, the **description**, and the **screenshot**. If the site name, device name, or location is missing, the immediate action is to **go back and ask the customer** for the necessary details. Tickets should not be left waiting for a day if crucial information is missing.
2. **L2 Escalation and Closure:** Issues should ideally be handled by the L1 support team or the Customer Success Management (CSM) team. Escalation to L2 support (e.g., to individuals like Dhana or the senior team) should only occur if the customer is **uncooperative** or fails to understand the resolution steps. The goal is for L1 support to **communicate with the customer and take the ticket to closure**.
3. **Consolidated Resolution:** When troubleshooting, resolutions should be **consolidated and solved for a complete, thorough resolution**, rather than solving problems as isolated "pockets".

4. **Customer Communication:** Customers like C-contact are generally expected to understand security issues and procedures. For certain operations, like retrieving historical playback, customers should be encouraged to use easy methods like providing a **Salvo link** related to the time of the incident. Any limitations regarding what can or cannot work must be **documented and published to the customer** to maintain clarity.

### B. Documentation and Knowledge Base Requirements

Maintaining accurate and comprehensive documentation is a continuous process essential for feeding the "Genie" (the AI knowledge base).

- **New Findings and Custom Parameters:** Any information regarding **custom parameters** or new solutions that is not currently present in the documentation must be **added as documentation**.
- **False Alarm Filtering Documentation:** A separate article dedicated solely to **false alarm filtering** should be created. This article must detail the necessary configurations required for the false alarm filtering mechanism to be enabled at the device level.
- **Port Information Detail:** Documentation must specify **what will happen if a specific required port is not enabled**.
- **AI Analytics List:** A comprehensive list of **all objects** that both current AI versions (e.g., Nova and the four identifiable objects) can detect must be compiled and documented.

### C. Advanced Analytics and False Alarm Classification

Genesis utilizes advanced AI analytics, which separates object identification from the alarm decision.

1. **AI Workflow:** The AI system operates in two distinct phases: **identification** (detecting objects like person, car, truck, etc.) and **decision** (classifying the event as a real or false alarm).
2. **Decision Parameters:** The decision (real/false classification) is entirely dependent on the **configuration parameters** defined in the priority list (custom properties). If an identified object (e.g., "person") is configured in the **priority list** as a target, the event is classified as a **real alarm**, even if the person is stationary.
3. **Troubleshooting AI Issues:** When an event detected by AI is incorrectly classified as a false alarm, the troubleshooting approach involves checking the three relevant custom parameters to see why the automated decision was made.

## VII. Detailed Device Integration Matrix

The following provides granular details on the integration, connectivity, and feature support for major device manufacturers supported by the Genesis platform.

### A. ADPRO (XO Client and NVRs)

ADPRO integration focuses on seamless alarm transmission via TCP connections and uses dedicated networking protocols for media streaming.

| Feature Category | Support Details | Source Reference |
|---|---|---|
| **Supported Modes** | Cloud Mode (Full support), Local Mode (Live Video, Playback, Local SDK Audio, Local PTZ, Local IO, Timeline) | |
| **Cloud Features** | Discovery, Live Video, Playback, Timeline, Events, ARM/DISARM, Genesis Audio (SIP), PTZ/Presets, IO (Input/Output), Event Acknowledgement, Play Audio from File | |
| **Integration Protocol** | Events are **pushed via TCP** to the dedicated Genesis ADPRO Receiver. Video streaming uses **RTSP** and **SDK image streaming**. | |
| **Networking/Ports** | Requires **RTSP Port 554**, **Control Port 2000**, and **Audio Port 3000** to be open at the customer site. The Genesis Cloud requires the **Genesis Adpro Receiver IP Address** (specific to the tenant) on **Port 10000**. | |
| **Identification** | Requires a **unique Server Unit ID** for each device for Genesis to accurately associate incoming alarms. | |
| **Prerequisites** | The ADPRO XO Client software is essential for setting up the device configuration. **ADPRO Analytics** (intrusion, motion, loitering traces) must be configured on the device to generate alarms. The `eventClipRecord` custom property may be used for automatic pre/post-alarm recording. | |
| **Limitations** | Genesis supports only **unencrypted strings**; the "encrypt string" option must not be selected. **P2P licenses** and **O keys** are currently **not supported**. | |

## B. Avigilon VMS

Avigilon integration primarily relies on VMS software installed on the customer's PC.

| Feature Category | Support Details | Source Reference |
|---|---|---|
| **Core Integration** | Genesis communicates with the Avigilon Control Center (ACC) server software. The process requires installing the **Avigilon webapi endpoint**. | |
| **Alarm Transmission** | For alarms to be transmitted, at least **one algorithm** (motion detection or analytics events) must be enabled on the camera or device level. | |
| **User Access** | A dedicated user account (e.g., "NXGEN") must be set up on the ACC client. This user must have **admin access/privileges**. | |
| **Network/Ports** | Requires opening the **Avigilon SDK HTTPS port** (Default **8443** for VMS and **443** for Avigilon appliance). | |

## C. Axis IP Camera and Axis Camera Station VMS

Axis devices rely on webhooks for event transmission and support advanced health features.

| Feature Category | Support Details | Source Reference |
|---|---|---|
| **Integration Protocol** | Uses **Axis HTTP API** and **WebHook Callback** for event transmission. Video streaming uses **RTSP**. | |
| **Event Configuration** | Event integration requires configuring a webhook on the IP Camera's rules to push video clips to Genesis. The webhook URL must be specified as `https://axisproxy.nxgen.cloud/eventIngest/<nxgenDeviceId>`. The `nxgenDeviceId` is used as both the **User Name and Password** for Basic Authentication. | |
| **Ports Required** | **Control Port** (Default **80** for IP Camera, **443** for Camera Station VMS), **RTSP Port 554**, and **Audio Port 3000**. | |

| | |
|---|---|
| **Features Supported** | Discovery, Live Video, Playback, Timeline, Events, ARM/DISARM, Genesis Audio (SIP), PTZ/Presets. Genesis supports all levels of Camera Health monitoring: **Basic, Basic+, and Advanced**. The system also supports pre/post-alarm recording integration (e.g., 5 seconds before and after the alarm). |
| **Features Not Supported** | Playing audio from a file is **not supported** for Axis devices. I/O functionality is planned for future support in Cloud Mode. Local Mode I/O and Local SDK Audio/PTZ/Timeline are currently under development. |

## D. Axxon VMS

Axxon employs a pull mechanism for events and mandates specific user permissions.

| Feature Category | Support Details | Source Reference |
|---|---|---|
| **Event Integration** | Events are **pulled by Genesis** via a **websocket subscription**; no manual configuration is required on the Axxon device side. | |
| **Supported Features** | Discovery, Live Video, Playback, Timeline, Events, ARM/DISARM, Genesis Audio (SIP), PTZ/Presets. Supports all levels of Camera Health. | |
| **Features Not Supported** | **I/O functionalities are NOT supported** for Axxon VMS integration. | |
| **Networking/Ports** | Requires the **Axxon HTTP(S) Port 80/443** to be open on the customer's firewall. | |
| **Prerequisites** | The user credentials provided to Genesis must have **Operator permissions**. **Strong passwords** should be implemented, though known issues suggest special characters like @ or # might not be supported. | |

## E. Camect AI Video Recorders

Camect is limited in its video archival features but supports core event functionality.

| Feature Category | Support Details | Source Reference |
|---|---|---|
| **Cloud Features** | Discovery, Live Video, Events, ARM/DISARM, Genesis Audio (SIP), PTZ/Presets, and Event Acknowledgement are supported. Live video is provided via an **HLS URL**. | |
| **Features Not Supported** | **Playback and Timeline are NOT supported** by the Camect device itself in Cloud Mode. Playback is also **not supported** in Local Mode. | |
| **Networking/Ports** | The Genesis Camect Receiver requires a specific IP address and **Port 10000** to be configured on the Camect device for alarm reception. | |

## F. Dahua NVRs and DoLynk Care

Dahua integration offers both cloud and local options, providing robust streaming and I/O support.

| Feature Category | Support Details | Source Reference |
|---|---|---|
| **Supported Modes** | Supports **Cloud Mode** (integration with Dahua DoLynk Cloud) and **Local Mode**. | |
| **Cloud Features** | PTZ Control, I/O Management, Arm/Disarm, Analytics (reducing false alarms by $\sim 80%$), Audio Integration (two-way communication via IP speakers), Device Health Monitoring. | |
| **Local Mode** | Supports operations for **P2P streaming, audio, and encrypted streams** on operator workstations. Requires installation of `Genesis-Installer.exe` (run as Administrator), Microsoft Redistributables, .NET Framework 4.8, and NodeJS (V18 preferred). | |

| | |
|---|---|
| **Prerequisites (DoLynk)** | Requires Product ID, Access Key, Primary Secret Access Key, Sub Domain Name, Device Model, and Box Credentials (Username and Password), along with the Category code (NVR, XVR, etc.). |
| **Networking/Ports** | Requires opening TCP, RTSP, and HTTP/HTTPS ports. For non-VPN setups, whitelist the Dahua Receiver IP **52.59.60.20**. |
| **Optimization** | **Smart Codec** support is conditional; enabling it may nullify certain network camera functions. **Time synchronization** must use `time1.nxgen.cloud` as the NTP Client server. |

## G. Ganz AI BOX

Ganz uses a webhook push mechanism and functions as an AI analytics device.

| Feature Category | Support Details | Source Reference |
|---|---|---|
| **Cloud Features** | Discovery, Live Video, Events, ARM/DISARM, Genesis Audio (SIP), PTZ/Presets, and I/O Management are supported. | |
| **Features Not Supported** | **Playback and Timeline are NOT supported** by the Ganz AI BOX itself in Cloud Mode. **Trigger health** is currently **not available**. | |
| **Integration Protocol** | Uses the **Ganz HTTP API and WebHook Callback**. Manual webhook configuration is required on the device. The URL uses the format: `https://ganzproxy.nxgen.cloud/eventIngest/<nxgenDeviceId>`. | |
| **Networking/Ports** | Requires opening HTTP/HTTPS (80/443) and **RTSP (554)** ports. NTP Client should be configured to `time1.nxgen.cloud`. | |

| | |
|---|---|
| **Optimization** | Configuration of **alarm rules is required** on the device, but this process is **prone to human error**. It is advised to configure specific analytics (intrusion, loitering) instead of generic motion detection to reduce alarm floods. For images, configure the device to send **multiple photos** (e.g., three) with a short interval (e.g., 500ms to 1 second) to aid Genesis analytics. |

## H. Hanwha-Techwin NVR

Hanwha relies entirely on Cloud Mode using webhooks for event reception.

| Feature Category | Support Details | Source Reference |
|---|---|---|
| **Cloud Features** | Discovery, Live Video, Playback, Timeline, ARM/DISARM, Genesis Audio (SIP), Events (specifically intrusion detection events). | |
| **Features Not Supported** | **PTZ and preset controls are NOT supported** in the current cloud integration. **Local Mode is NOT supported**. | |
| **Integration Protocol** | Events are transmitted via **webhooks**. The webhook URL must be configured on the camera rules within the Hanwha device. | |
| **Networking/Ports** | Whitelist the Genesis Alarm Receiver Gateway IP **18.184.110.24**. Requires opening HTTP/HTTPS (80/443) and **RTSP (554)**. | |
| **Prerequisites** | HTTP or HTTPS port and RTSP port are mandatory fields to provide during configuration. | |

## I. Heitel (Legacy Device)

Heitel devices require specific port access and often manual processing due to their legacy nature.

| Feature Category | Support Details | Source Reference |
|---|---|---|
| **Cloud Features** | Discovery, Live Video, Events, ARM/DISARM, Genesis Audio (SIP), PTZ/Presets, and I/O Management are supported. | |
| **Features Not Supported** | **Playback** is **not supported** by the device itself in the context of Genesis integration. | |
| **Integration Protocol** | The device must be reachable via the **CamControl Server**. Alarm reception often requires **raising an individual ticket** on the NXGEN help-desk portal for manual intervention due to Heitel's **legacy nature**. | |
| **Networking/Ports** | **Port 3333** must be open and configured on the HTConnect Tab of the Heitel device. Requires whitelisting Genesis IP addresses. | |
| **Prerequisites** | Requires the device's public static IP address, Serial Number, Control Port, and RTSP Port. User account must have **"Operator permissions"** at a minimum. | |

## J. Hikvision and HikProConnect

Hikvision integration relies heavily on time synchronization and the use of cloud base URLs for HikProConnect.

| Feature Category | Support Details | Source Reference |
|---|---|---|
| **Integration Protocol** | Event integration uses **ISAPI Event API Subscription**; no specific device configuration is needed on the device beyond enabling "Notify Surveillance Center". | |
| **Cloud Features** | Device Discovery, Event Reception/Management, Alarm Processing/Analytics, Arm/Disarm, Live Video, Playback, PTZ/Presets Control, I/O Management, Genesis Audio (SIP), Timeline. | |

| | |
|---|---|
| **HikProConnect Setup** | Requires the **Device Base URL** (e.g., `https://ieu.hik-partner.com` for Europe) in the "IP address-host" field. The **Encryption Key** is required, but its functionality is **only supported when Local Mode is installed**. |
| **Prerequisites** | Requires an Admin user with specific permissions (Live View, Playback, PTZ control, Notify Surveillance Center, Time Zone Sync). The device must be configured with **continuous recording** (or Motion&Alarm with 30 seconds pre/post-record minimum). Requires the **unique Server Unit ID**. |
| **Networking/Ports** | Requires whitelisting the Hikvision Receiver IP **35.156.60.98**. Requires TCP, RTSP, and HTTP/HTTPS ports. |
| **Time Synchronization** | Device time must **precisely match** the time configured in Genesis. Configure the NVR to use **`time1.nxgen.cloud`** as its NTP client server. |

## K. Milestone VMS

Milestone requires extensive port opening and high user privileges.

| Feature Category | Support Details | Source Reference |
|---|---|---|
| **Cloud Features** | Discovery, Live Video, Playback, Timeline, Events, ARM/DISARM, Genesis Audio (SIP), PTZ/Presets (including Milestone-specific UI enhancements like Rectangle PTZ), I/O Management. | |
| **Features Not Supported** | **Advanced Camera Health monitoring is NOT supported**. Event Acknowledgement and Play Audio from File are **not applicable/supported**. Local IO is currently under development. | |

| | |
|---|---|
| **Networking/Ports** | Requires opening specific ports: **443** (Management Client), **7563** (Streaming), **8081** (Web Client), and **22331** (Events). Whitelist the Milestone Alarm Receiver Gateway IP **3.66.98.181**. |
| **Prerequisites** | The user must be able to log in to the Milestone management client using **basic or Windows authentication**. The user account provided to Genesis **must have admin privilege**. Servers require **Microsoft .NET 4.7.2 Framework and Microsoft .NET 6 Runtime** installed. |

## L. Reconeyez PIR Cam

Reconeyez operates as an intrusion panel, necessitating a complex alarm flow routed through Talos.

| Feature Category | Support Details | Source Reference |
|---|---|---|
| **Device Representation** | Represented as a **single camera device** in Genesis. | |
| **Alarm Flow (Dual Path)** | **Raw alarm events** are sent from Reconeyez to **Talos via SIA DC-09**. **Alarm images** are sent simultaneously to a **Genesis FTP**. Talos then converts the C-alarm message to a Genesis alarm, which Genesis processes, pulls images, applies analytics, and sends a follow-up alarm back to Talos. | |
| **Supported Features** | Events/Alarms reception and ARM/DISARM. | |
| **Features Not Supported** | Due to its battery-operated nature, **Live video, Playback, and Timeline are NOT supported**. Local Mode operations are also not applicable. | |

## M. Senstar VMS (Senstar Symphony Server)

Senstar integration relies on consistent event polling and does not support I/O or Local Mode.

| Feature Category | Support Details | Source Reference |
|---|---|---|
| **Integration Protocol** | HTTP API with **Event Polling**, where Genesis periodically requests event data from the Senstar Symphony Server. | |
| **Core Capabilities** | Discovery, Live Video (using RTSP), Playback, Events, ARM/DISARM (software arm), Genesis Audio (SIP). Health polling from the cloud is supported. | |
| **Features Not Supported** | **Input/Output (I/O) functionality is NOT supported** by the Senstar device integration. **Local Mode** operations are **NOT supported**. Timeline is only partially supported. | |
| **Prerequisites** | Senstar Symphony Server must be installed and operational. Requires a unique **"Unit ID"** for integrations involving the Senstar Network Manager. | |
| **Networking/Ports** | Requires Mobile API ports (8488) and RTSP port. Time synchronization must use `time1.nxgen.cloud`. | |

## N. Viasys Cloud NVR

Viasys uses a unique placeholder configuration process.

| Feature Category | Support Details | Source Reference |
|---|---|---|
| **Integration Steps** | Requires creating a **placeholder (dummy) device** in Genesis first. The Genesis **Device ID (GUID)** is then shared with the Viasys team, who handles the configuration of physical sensors on their end. | |
| **Cloud Features** | Discovery, Live Video, Playback, Timeline, Events, ARM/DISARM, Genesis Audio (SIP), PTZ/Presets, and **I/O functionalities** are all **supported** once Viasys completes the configuration. | |

| | |
|---|---|
| **Integration Protocol** | Uses the **Viasys HTTP API** and a **webhook callback mechanism** to push events to Genesis. |

### O. Other Devices (NetVue, Uniview, InnoVi)

- **NetVue IP Camera:** Integration uses an **HTTP API with Event Polling**. Requires specific port configurations, including RTSP Port **8488**. The password requires encryption via the Senstar Password exe for security.
- **Uniview NVR:** Uses Uniview C# SDK and .NET SDK Subscription for events, implying a push mechanism. Requires **Operator permissions**. Supports all levels of Camera Health (Basic, Basic+, Advanced) and features including Live Video, Playback, Timeline, Events, ARM/DISARM, Genesis Audio (SIP), PTZ/Presets, and I/O.
- **InnoVi AI Cloud:** Supports **Discovery** and **Events** (Metadata & Event data) via an API Token and HTTP API. The AI focuses on core analysis and event generation. Like other AI boxes, direct Live Video streaming and Playback are typically sourced from a connected camera/VMS, **not the InnoVi device itself**.

## VIII. Network and Security Configuration (IPs and VPNs)

Robust security requires precise network configuration and IP whitelisting.

### A. NXGEN Public Gateway and Messaging IP Addresses

For non-VPN customers, the following core public IP addresses must be whitelisted on the customer's network/firewall to ensure Genesis can communicate with on-premise devices:

| Service | Public IP Address (Primary) | Public IP Address (Secondary) |
|---|---|---|
| **Genesis Services Primary Gateway** | **18.185.17.113** | **3.124.50.242** |
| **Streaming Gateways** | **3.126.237.150** | **3.75.73.51** |
| **Genesis Messaging Services** | **3.127.50.212** | N/A |

### B. Device-Specific Receiver IP Addresses (Non-VPN)

In addition to the core gateways, specialized receiver IPs are required for certain devices (which must be whitelisted):

- **Dahua Receiver:** 52.59.60.20.
- **Hikvision Receiver:** 35.156.60.98.
- **Hanwha Alarm Receiver Gateway:** 18.184.110.24.
- **Milestone Alarm Receiver Gateway:** 3.66.98.181.
- **ADPRO Receiver:** Tenant-specific IP on Port 10000.
- **Camect Receiver:** 3.122.169.231 (in addition to gateways).

## C. VPN Connectivity Options

VPN connectivity is the preferred and more secure method for connecting customer networks to Genesis, as it allows internal IP addressing to be used. NXGEN offers three main VPN options based on the customer's router compatibility:

1. **Primary Option: OpenVPN:** Highly recommended if the customer's router supports the OpenVPN client configuration. NXGEN provides a custom `.ovpn` configuration file for easy import. Requires the **Customer LAN IP Segment**.
2. **Secondary Option: IPsec (IKEv1 or IKEv2):** Used for routers that support IPsec site-to-site VPNs. Requires the **Customer Router Public Static IP**, **Router Model**, and **Customer LAN IP Segment**.
3. **Optional Option: WireGuard VPN:** A modern, high-performance protocol offered if the customer's router supports the WireGuard protocol.

## D. General Port and Time Requirements

- **P2P/Websockets:** For P2P streaming (often used in Local Mode for Dahua and other devices), inbound websockets must be whitelisted on port ranges **10000 to 10500** and **14890 to 14990**.
- **NTP Server:** Time synchronization is critical for event correlation. All devices (including Dahua, Hikvision, Ganz, Senstar) should configure their NTP Client to use **`time1.nxgen.cloud`**.

---

In essence, the entire NXGEN system operates as a **Universal Translator Hub**. Devices, regardless of their native complexity (like the dual-protocol flow of Reconeyez or the software-centric nature of Milestone VMS), speak their local "language" to a dedicated **microservice proxy**. This proxy instantly converts the incoming data into the **standardized Genesis/Talos format** (JSON/API calls). This robust, layered architecture—reinforced by mandatory IP whitelisting and precise time synchronization—ensures that the central operator (Talos) receives and acts upon consistent, verified alarm data, even though the underlying technology stack is highly varied.

This comprehensive article synthesizes all available information from the provided documents, encompassing security industry standards, market data, and detailed operational

troubleshooting training transcripts. The structure is designed to compile all knowledge points into a single, cohesive resource for an AI tool.

---

# The Unified Knowledge Base: Security Platform Operations, Troubleshooting, and Industry Standards

## I. Introduction and Organizational Training Mandate

This document serves as a comprehensive knowledge base, compiling essential information regarding organizational structure, platform operations (Genesis and Tallos), detailed troubleshooting methodologies for Level 1 (L1) support, device integration standards, and the broader security market landscape.

### The Mandatory L1 Training Program

A **mandatory training program** was established to ensure all team members are hands-on in handling L1 support and onboarding processes. The training is scheduled for one month, with a two-week checkpoint to evaluate progress and modify the approach as needed. **Full attendance** is required, and schedules are to be aligned accordingly, modifying timing only if pre-existing, unchangeable conflicts exist.

The primary goal of this initiative is to **close the knowledge gap** across the organization. This training is not limited to the core group; Dana SHA from the Quality Assurance (QA) team will also participate. Furthermore, team members like Rajesh or Vignesh who possess unique knowledge of certain modules or tickets are expected to provide training to the entire group. The target for this intensive training period is the current month or the next four weeks.

### Roles, Responsibilities, and Support Structure

#### Customer Success Managers (CSMs) and L1/L2 Support

The CSM team is expected to **act as L1 support** to resolve issues at their level. This includes handling the customer directly and updating the ticket analysis part. The ultimate goal is to **reduce the time the customer waits for a response and resolution**.

- **L1 Responsibilities (CSM Role):** The CSM is the **owner of the task** and the primary contact for the customer. They should aim to resolve issues within **24 hours** for the

analysis part. If an issue is confirmed to be a bug or requires development work, it must be passed on to the development team, but only after thorough analysis. When resolving tickets, any custom parameters or configurations discovered that are missing from documentation must be added to the documentation.

- **L2 Support:** L2 support (Mehana or Yan) works on tickets if L1 (CSM) lacks the bandwidth due to meetings or other conflicts. L2 resources are also available to support L1 staff (e.g., if help is needed, one CSM can reach out to others or to L2 members like Dana and Yan).
- **Escalation Criteria:** A ticket should not come to L2 unless L1 is busy or if the problem goes beyond L1 observation. If troubleshooting takes too long (e.g., more than 30 minutes during a regular day), it should be forwarded to L2. Complex or non-obvious issues requiring extensive checks should also be escalated.
- **Team Operation:** The team is expected to **support each other**. CSMs should proactively check the CSM ticket queue and pick up unassigned tickets, regardless of who the customer owner is, especially if others are busy or on leave.

**Documentation and Knowledge Management**

Documentation, particularly regarding custom properties and troubleshooting steps, is crucial. Team members have the rights to **create and update documentation**.

- **Custom Properties Documentation:** Custom properties related to false alarm filtering (priority list, whitelist, blacklist) currently lack a dedicated, complete article. A separate article only for false alarm filtering configurations should be created, detailing step one (device-level filtering enabled) and step two (checking custom properties like blacklist criteria).
- **Guidelines for Documentation:** Documentation, especially troubleshooting guides, should be **precise, small, and to the point**, possibly using screenshots or flowcharts, avoiding excessive verbosity. The documentation needs to capture the impact of not implementing a certain setting (e.g., if a port is not enabled, what functionality fails).
- **Device Parameters:** A consolidated, internal sheet or document should be maintained for all custom properties required for different device integrations (e.g., Milestone, Hikvision, Eagle, Senstar, AJAX, etc.) to avoid checking multiple documents or logging into different tenants every time. This information should not be shared externally.

# II. Platform Architecture, Troubleshooting Fundamentals, and Alarms

## The Genesis-Tallos Ecosystem

The platform relies on Genesis for video analytics and Tallos (or AMVIN) for receiving and processing alarms.

- **AI Workflow (Old AJAX Example):** In older implementations, the alarm initially comes to **Tallos**, which forwards it to **Genesis** for AI processing. Genesis then performs the AI analysis and **resends the alarm back to Tallos**. This workflow involves receiving a BA alarm in Tallos, triggering a workflow to get three images, calling the N extent false alarm filtering API, and returning the result. If the result is *true*, it goes to the operator; if *false*, the alarm is discarded or closed.
- **Alarm Routing (Lindhorst Case):** Routing alarms requires a two-level setup, one in Genesis and one in Tallos. This involves setting up **Customer Groups** in Genesis to limit operator visibility to specific sites. In Tallos, a similar structure must be aligned, including creating **Organisation Units** and **Alarm Channels** to ensure operators only see alarms from sites they have access to. Users are allocated to a specific **Workspace** which bundles the allowed alarm channels.

## Network Connectivity and Access

Devices can be onboarded via public IP, VPN, or **white listing**.

- **White Listing (Jump Server):** If a public IP or VPN is not used, white listing is configured, meaning only a predefined list of IPs (e.g., the engine IPs or proxy IPs, such as the event processor or Dahua proxy) are allowed access to the customer's network via router or firewall settings. To access a white listed device, users must remote into a **shared virtual machine** (or "jump server"). Due to this being a shared resource, caution is needed to avoid terminating another session, and critical access must be communicated via chat.
- **IP/Port Requirements:** For standard integrations like Hikvision (NVR type), three ports are ideally needed, along with the IP, username, and password. Cloud-based platforms (Hit Pro Connect, Dahua Cloud, Access Camera Station) do not typically use IPs in the same way.

## Device States and Alarms

### Arm/Disarm vs. Isolate

These states control alarm processing and are often manually triggered or automated via workflow APIs.

- **Arm/Disarm:** When a device is disarmed, the state is continuous until manually armed back, and vice versa. This is typically used when the user has a process to arm it back.
- **Isolate:** The mandatory parameter for isolate is a **time duration** (e.g., 30 minutes default). When a device is isolated, alarms are received, but the system **will not fetch the image, apply AI, or forward the alarm**. Isolating means the alarm will not be processed.

### Alarm Blocking and Event Overflow

Alarms can be blocked if they exceed a specific threshold, typically indicating a device problem.

- **Overflow Threshold:** The default threshold is **25 alarms** in a 5-minute window if no custom property is configured. If the alarm count exceeds this (e.g., 46 alarms received against a 25 alarm threshold), the system blocks subsequent alarms. This must be communicated to the customer as a device problem.
- **Customizing Thresholds:** The threshold can be configured via a specific custom property (found in CLUE or additional settings, near AI filtering). If the system blocks alarms below a customized threshold (e.g., blocks at 46 alarms when configured for 50), it is a Genesis mistake.
- **Troubleshooting Blocked Alarms:** The first check involves looking at the main dashboard and the device-specific dashboard, filtering alarm logs, focusing on Analytics motion alarms, and checking for blocked/redundant/overflow events. Tallos has its own blocking logic, applying at the site level, regardless of the device count; if more than 25 alarms occur in a 5-minute window, Tallos may block them even if Genesis forwarded them.
- **Customer Recommendation:** For devices like Axon, which tend to send too many alarms and frequently get blocked, customers should be asked to change configuration from simple **motion alarms** to **IVS alarms (Intelligent Video Processing alarms)**, which include a level of AI filtering applied by the device itself before transmission.

# III. AI and False Alarm Filtering Deep Dive

Artificial Intelligence (AI) is utilized in security for real-time data analysis, distinguishing genuine threats, and reducing false alarms.

## AI Classification Mechanisms

The AI process is divided into two parts: **identification** (identifying objects, independent of configuration) and **decision** (deciding if the event is a real or false alarm based on configuration parameters). AI detection operates without exclusion rules, meaning the algorithm attempts to detect everything.

Custom properties dictate the AI’s decision:

- **Priority List:** If a detected object (e.g., "person") is configured here, it is treated as a **100% real alarm**. For a priority item, classification as a real alarm occurs even if the object is static (not moving).
- **Whitelist:** A detected object on the whitelist (e.g., "car") has a **potential to be a real alarm**.
- **Blacklist:** Any detected object configured here (e.g., "bicycle") is flagged as a **false alarm** (i.e., suppressed).
- **Reject Unknown:** This setting deals with objects that cannot be identified (classified as "unknown" or "NA"). If only an unknown object is detected (e.g., due to wind movement

or light reflection), the alarm should typically be **suppressed as a false alarm**. The inclusion of "NA" (Not Applicable/Not Classified) in this rejection bucket requires discussion with development (ASR).

## Troubleshooting Incorrect Classification

If alarms are incorrectly classified (real vs. false), the primary check is the **Analytics dashboard** to ensure false alarm filtering is applied. If filtering is applied, the next steps are to check the custom properties:

1. **Review Custom Properties:** Check the four custom properties (priority list, whitelist, blacklist, etc.) to see if the incorrectly flagged object (e.g., "person") is marked as a rejection/blacklist criteria.
2. **Check for Masking:** Verify if the area generating the alarm was **masked** in the configuration. Masking might not always be indicated by a red region.
3. **Data Analysis and Escalation:** If settings appear correct but misclassification persists (e.g., a person is detected but marked as a false alarm because "person" was missing from the priority list), documentation must be prepared and escalated to the data science team (Nazem or Asan).

**Critical Documentation Gap for AI:** There is currently no finite list of classifications that the AI versions (V4 and Nova/V6) support (e.g., if "person" is detected, does the AI return "person," "human," "pedestrian," "man," "woman," etc.?). A comprehensive list of all supported tags must be obtained from the development team to ensure accurate custom property configuration.

## Ticket Analysis Examples: AJAX (4796) and C Contact (4792)

### AJAX Filtering Issue (Ticket 4796)

The issue reported was **no filtering** for AJAX sites at the customer level (Securitas Pain). Initial analysis showed AI was, in fact, working at a high level (person classified). The customer had a large number of sites.

- **Old Implementation:** This customer uses the **older way of implementing AJAX** via a workflow. This setup is highly cumbersome, requiring extreme expertise in **Tallos and Genesis** for troubleshooting. Only a few individuals (Rajesh and the speaker) can figure out this implementation.
- **Migration Recommendation:** Due to maintenance problems and complexity, new AJAX customers are not onboarded this way. Discussion should occur with Eve (EV) regarding migrating Securitas Pain sites to the **Genesis AJAX** implementation, despite their strict quality procedure for testing.
- **Device Type:** Securitas uses PIR CAMs (passive infrared cameras) that take three or four images when an alarm from the PIR CAM is received, and these images are analyzed by Genesis. The images do not come from a traditional camera installed on AJAX.

- **Configuration Check (Generic):** Normally, if AI is reported as not working, the first step is to check the Analytics dashboard to see if **False Alarm Filtering is turned on**. For this specific, older workflow implementation, however, the AI appears to work even if False Alarm Filtering is *not* turned on, because the system is calling AI through an API via the workflow, bypassing the usual Genesis configuration checks (a known issue).

**C Contact Alarms Blocked Issue (Ticket 4792)**

The customer reported alarms were blocked at site 80091.

- **Device Context:** The site has two devices, including an **Axon** device. Axon devices are known to send too many alarms and frequently get blocked.
- **Overflow Cause:** The system's threshold is 25 alarms (default). The system received 46 alarms in the defined period, requiring the system to block them. This is considered a **device problem** that the platform will not support, and the customer should be told to tune their analytics.
- **Troubleshooting Steps:**
    1. Check if alarms are blocked currently (as of the check date).
    2. Check the event overflow logs for the site.
    3. Check the custom property related to the blocked alarm overflow threshold to confirm if it is set above 25.
    4. If Axon is the cause, recommend configuring **IVS alarms** instead of motion alarms to introduce a level of filtering before alarms reach Genesis.

# IV. Device Integration and Troubleshooting Case Studies

## Dahua Cloud Integration and Discrepancy (Term Watcher)

A customer reported thousands of alarms in Genesis (2700) where the physical device log showed very few (8-10 intrusion alarms).

- **Alarm Multiplicity:** This discrepancy suggests Genesis is interpreting every Dahua alarm as 10 or 100 alarms (a huge gap).
- **Troubleshooting Process (Establishing Problem Source):**
    1. Access the device (often requiring a jump server for white listed environments).
    2. Check the device log and compare counts with the Genesis dashboard and the **alarm receiver log**.
    3. Install the **Dahua SDK test client** (downloadable from the internet, e.g., Dahua Partner Alliance) on the jump server.
    4. Input the device IP/Port and credentials into the test client.
    5. Compare the test client's received alarm count to the device log count.
        - If the test client count matches the device log (e.g., 8 alarms), the issue is with **Genesis**.

        - If the test client count matches the Genesis dashboard (e.g., thousands of alarms), the issue is with **Dahua**.
- **Ownership and Escalation:** If it is a Genesis problem, pass it to the tech team. If it is a Dahua problem, communicate the discrepancy to the customer and reach out to Dahua support. Ideally, the tech team handles vendor communication as logs and traces might be needed.

## Hikvision (Hi-Vision) and True Vision

Hikvision presents multiple flavors of integration (NVR, IP camera, hybrid) due to network configurations and device restrictions.

### Hikvision Onboarding and Configuration

- **NVR vs. IP Camera:** NVRs (Network Video Recorders) handle processing and connect multiple cameras (e.g., 16 or 32 channels). IP cameras are single devices, sometimes with two lenses.
- **Cloud vs. Local Mode:**
    1. **IP Camera Recommendation:** Use **Cloud Only mode** to centralize streaming requests through a single Genesis local mode connection, preventing bandwidth issues caused by multiple direct connections to the device (which has less processing capacity).
    2. **NVR Standard:** Normally, NVRs use local mode, but hybrid configurations (e.g., V-ZAG/HESTA) must use **Cloud Only mode** if audio transmission is blocked by local mode installation or firewall settings.
- **Device Setup Checklist (Standard NVR):**
    1. **IP Reachability:** IP must be reachable from the Genesis machine, or via a jump server if whitelisted.
    2. **Ports:** Ideally, three ports are needed.
    3. **Time Zone/DST:** Time zone settings must match the device, and **Enable DST** (Daylight Saving Time) must be active for automatic adjustments.
    4. **Recording:** Continuous recording is preferred. Event-based recording is an alternative, often recommended at **30 seconds** duration.
    5. **Event Linkage:** When alarms are configured on the device, the trigger channel must be **linked/associated** to the camera channel ID so Genesis knows the source of the alarm.
    6. **Alarm Protocol:** Hikvision and Genesis talk using the **Do HTTP(S) request** protocol for events. The request must include the Genesis event ID (which is the NX Witness camera ID).

### True Vision Specifics

True Vision devices are products from Hikvision but use a different UI and SDK.

- **Onboarding:** True Vision devices are onboarded by selecting the **Hik Vision** device type in Genesis.
- **Polling vs. Events:** Unlike standard Hikvision (which normally sends events), True Vision integration uses **polling** for event fetching, implemented by Rajesh. Avigilon is the only other device type noted to use polling.
- **Alarm Mapping:** On True Vision devices, all **IO (Input/Output) alarms** are currently mapped as **analytics.intrusion alarms** on Genesis. This is crucial for troubleshooting data reconciliation.
- **Troubleshooting Gaps:** True Vision alarms are often **not recorded in the Alarm Receiver Log**. This is considered an integration gap that must be raised to the development team.

## General Device Troubleshooting Steps

| Issue Category | Initial Checks (L1 Analysis) | Advanced Checks / Escalation Trigger |
|---|---|---|
| **No Events from Single Camera** | 1. Check ticket for site, device, and sensor names. 2. Check the overall Dashboard and the Device Dashboard for the specific sensor. 3. Check the **Alarm Receiver Log** (if empty for that camera). | 4. Log into the device UI to confirm alarms are present there. 5. Check device configuration for "Notify Surveillance Center" or equivalent setting (e.g., "Event Linkage") to ensure alarms are configured to be sent to Genesis. |
| **No Image / Live View Disabled** | 1. Check **Video Activity Search**: Are images available for alarms?. 2. Check device UI for Live View/Playback: Is it working on the device itself? (If not, customer issue). 3. If images/live view fail, check **Encoding/Stream Setting** (e.g., H.265 requires specific CODEC installation; downgrade to H.264 if needed). | 4. Check **Genesis Web Client, Device Web Client, and Test Client** playback functionalities. If it works on test client but fails on Genesis, it is a Genesis issue. 5. If specific features (like audio) fail in local mode, recommend switching to **Cloud Only mode** (especially for Hikvision). |
| **Alarms Present but Not Forwarded to Operator Screen** | 1. Check the Device Dashboard's **Genesis Analytics** column. 2. If all inputs are *false*, the alarm will not be forwarded. 3. Check if the device is **disarmed** or **isolated** (will be visible in Genesis Analytics column). 4. Check the `report alarm` toggle. | If configuration is armed/active but alarms are still suppressed, review **False Alarm Filtering** (Priority/White/Blacklists). |

# V. Security Standards and Protocols

## European Standards

### EN 50518: Monitoring and Alarm Receiving Centre (MARC)

EN 50518 is the **European certification standard** for Monitoring and Alarm Receiving Centers (MARC). An EN 50518 certificate has European validity.

- **Scope:** The standard encompasses a wide range of certification processes, covering both security and non-security applications.
- **Main Functions of a MARC:**
  - **24/7 availability**.
  - A handling process that meets the **required response times**.
- **Usability:** It is widely used for centers handling messages from: intrusion/hold-up alarms, video surveillance, access control, fire detection, social alarms, temperature monitoring (cold storage), transport monitoring, and infrastructure projects (e.g., bridges, tunnels, power plants).
- **Certification:** Certification requires assessment by an accredited certification body. The process includes defining category and scope, leading to an independent acceptance audit.

### VdS 2465-4en (S6/S7 Protocol)

The VdS guidelines define a manufacturer-independent interface for the **uniform transmission of signals/messages** from the Alarm Receiving Centre (ARC) to an Alarm Service (AS) via TCP/IP. These guidelines are valid from March 1, 2021.

- **Protocol Mechanism:** Transmission between the ARC (alarm provider function) and AS shall be realized via **TCP/IP connections**. The **S6 interface shall operate as a TCP/IP server**, and the **S7 interface as a TCP/IP client**.
- **Transmission Paths:** At least two physically separated transmission paths must be available between ARC and AS, and their availability must be monitored.
- **Data Types and Requirements:**
  - **MUST Elements:** Required in the minimum scope of a protocol implementation, but not necessarily transmitted with every signal/message.
  - **DUTY Elements: Shall be transmitted with each signal/message** (minimum extent of a signal).
  - **CAN Elements:** Optional and may be discarded if not supported by the receiver's protocol implementation.
- **Additional Information:** Additional information (e.g., video pictures or audio streams) are distinguished from processing signals. If a video image *is* the message requiring processing (e.g., motion detection), the ARC must generate a message forwarded via S6/S7, but the image itself can reach the AS by other means (bypassing the ARC).
- **Security:** Confidentiality is ensured using the **ARC-256 encryption algorithm**.

### BS 8418:2015 CCTV Systems

This British Standard provides a checklist and requirements for CCTV Systems, particularly for remote video monitoring.

- **Planning and Design:** Requires completion of a **threat assessment and risk analysis** prior to design, and the creation of an operational requirement document addressing customer needs.
- **Police Response:** CCTV systems **without a Unique Reference Number (URN) will not be given Level 1 Police response**.
- **Compliance:** Installers must adhere to the Data Protection Act 1998, the ICO CCTV code of practice, and the Surveillance Camera Commissioner's code of practice.

## Digital Communication Protocols

### Ademco Contact ID Protocol

This is a digital communication standard developed by the ADEMCO Group for alarm system communications (SIA DC-05-1999.09).

- **Mechanism:** Utilizes **standard DTMF tones** for information transmission.
- **Session Components:** A communication session consists of three basic elements: the Handshake Tone sequence, Message Blocks, and Acknowledgements.
- **Handshake Tones:** Produced by the RECEIVER to signal the TRANSMITTER that the communication channel is ready.
- **Goal:** To provide across-the-board compatibility for equipment designed to this standard, regardless of the manufacturer.

### SIA DC-09-2021 (IP Event Reporting)

This standard (Digital Communication Standard – Internet Protocol Event Reporting) was published in 2021 by the Security Industry Association (SIA). Compliance is voluntary, and it is intended for use by manufacturers of control panels and central station receivers to ensure equipment compatibility.

# VI. Security Industry Market and Technology Trends

## Market Growth and Trends

The alarm monitoring industry, particularly in Europe and Latin America (ELA), is experiencing significant growth and technological transformation.

- **ELA Market Forecast:** The ELA alarm monitoring market size was valued at **$7.33 billion in 2020** and is projected to reach **$21.66 billion by 2030**, registering a CAGR of 11.5% from 2021 to 2030.

- **Drivers:** Growth is primarily driven by the **rise in demand for AI-powered home automation solutions** and the increasing popularity of smart homes, especially in the UK, Germany, and Brazil.
- **Segment Performance (2020):**
    - **Systems & Hardware** accounted for the maximum revenue.
    - **Analog Signal and Protocol Signal** segments collectively held 72.1% market share.
    - The **IP-Network segment** is projected to show the highest growth rate (CAGR of 13.9%).
- **Geographic Focus: Brazil** contributed a major share (more than 14.3% in 2020) in the ELA market. The German Home Security Systems market is also seeing robust growth due to rising concerns over residential security and technological advancements (IoT, cloud-based monitoring).

## Technological Transformation

The UK alarm signalling market is rapidly transitioning from copper-based PSTN lines to a fully digital communication network, scheduled for completion by 2027.

- **Digital Infrastructure:** The shift to digital communication (VoIP, fiber-optic, 5G) enables faster, more secure, and future-proof solutions.
- **Key Innovations:**
    - **Artificial Intelligence (AI):** Reshaping alarm management by allowing faster, more accurate responses and differentiating genuine threats from false alarms.
    - **IoT Integration:** Allows alarm systems to communicate seamlessly with smart devices (e.g., activating lighting, adjusting surveillance angles) to secure a property in real time.
    - **5G Connectivity:** Redefining operations through low latency and high-speed data transfer, enabling instant transmission of high-quality video and data for real-time monitoring.

## Industry Players and Ecosystem

Key industry players profiled in the ELA alarm monitoring market include ABB Ltd., Siemens AG, Diebold Inc., Schneider Electric SE, Honeywell International, Vivint Inc., Johnson Controls, Bosch Rexroth AG, ADT Corporation, and Vector Security.

The software ecosystem features platforms like **Immix**, a provider of commercial central station and remote SOC software for managed video and security services. Immix has integrations with various partners:

- **Trackforce:** Integration with TrackTik software enhances remote guarding automation and security event management.
- **PureTech Systems:** Integration of PurifAI (geospatial AI-boosted video analytics) with IMMIX CS & GF platforms significantly reduces nuisance alarms.

- **Hanwha Vision:** Enhanced partnership supports industry-pioneering AI analytics in video surveillance for Alarm Receiving Centres (ARCs).
- **Eagle Eye Networks:** Integration connects the Eagle Eye Cloud VMS (video management system) with Immix central stations, leveraging cloud advantages like centralized management, cybersecurity, and AI-filtering.
- **Arcules:** Integration provides Central Station and Guard Force users with tools to manage, verify, and respond to security events using Arcules-managed video feeds.
- **Ajax:** Integration of Ajax alarms into the IMMIX CS and GF platforms.
- **Ramtech:** Partnership integrates Ramtech's WES wireless evacuation system with the Immix platforms to provide a combined fire protection and safety solution.
- **VIVOTEK:** Partnership provides Immix customers access to VIVOTEK NVRs (network video recorders) and audio/video management solutions.